

Roof Lifecycle Cost Analysis

Culbertson (N. Virginia) market
October 2, 2026

Planned maintenance is projected to save the owner of a typical Culbertson (N. Virginia) school **\$1.62M** over the next 20 years.

That is 37.0% less than fixing problems only after they leak. Under conservative to upside assumptions, the savings range from \$851K to \$2.11M. In today's dollars the estimate is worth \$843K.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE	NEW ROOF TODAY
Culbertson (N. Virginia)	K-12	120,000 sq ft	Mod bit	8 yrs Good	\$1.14M

PAXSEAL NET COST \$2.76M \$2.65M spent plus roof wear, net of roof life left	REACTIVE NET COST \$4.38M \$3.85M spent plus roof wear, net of roof life left	EMERGENCY COSTS CUT \$1.88M Unbudgeted repairs and storm damage avoided	NEW ROOF PRICE 2× in 15 yrs \$1.14M today, \$2.8M by year 20
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Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal program fees	\$441K	—	-\$441K
Repairs paid by owner	\$278K	\$2.13M	+\$1.85M
Expected storm damage	\$57K	\$86K	+\$29K
Roof replacements	\$1.87M	\$1.63M	-\$236K
Current roof value used	\$684K	\$684K	—
Credit for roof life left	-\$570K	-\$152K	+\$418K
Net 20-year cost	\$2.76M	\$4.38M	+\$1.62M

What this means

- 1 **A later replacement.** The next new roof moves from year 8 to year 11, when it costs less in today's dollars.
- 2 **Smaller repairs.** Fixes made on scheduled visits avoid emergency work at about 3.9× the cost in a school, saving \$1.85M.
- 3 **A predictable budget line.** Emergency repairs become a fixed annual cost the board can plan for, and leaks stop closing classrooms.
- 4 **The program pays for itself.** Each \$1 of fees avoids about \$4.68 of other roof costs.

Local risk profile

Prince William County, VA (51153)

Thunderstorm wind / derecho, snow and ice

Strong wind and winter weather both rated Relatively High (FEMA NRI); tropical remnants ~1 per 12 years. Expected storm damage to this roof over 20 years is **\$57K** if maintained and **\$86K** if neglected. A documented maintenance history also helps when an insurer questions a claim.

Hail	Moderate	
Strong wind	High	
Tornado	Moderate	
Hurricane	Moderate	
Winter weather	High	
Ice storm	Low	
FEMA National Risk Index rating		

Key assumptions

The basis for each figure in this analysis.

ASSUMPTION	VALUE	BASIS
Roof life	20 / 15 yrs	Maintained / reactive. Fannie Mae service-life tables; +5 yrs per NRC Canada and warranty terms
Roof cost growth	4.6%/yr	BLS Producer Price Index, nonresidential roofing contractors, 2007–2025
Replacement cost	\$9.50/sq ft	Culbertson (N. Virginia) contractor price ranges
Reactive repair cost	3× planned × 1.3	Industry estimates; no primary study found. ×1.3 for leak damage in a school
Storm damage	0.15%/yr	Of roof cost, before insurance. FEMA National Risk Index expected building loss, roof-related hazards
PaxSeal fee	\$16,400/yr	PAX program estimate for this roof size. Your proposal will state the actual fee

Estimates, not a quote. Projections from Culbertson (N. Virginia) market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20. Storm damage is before insurance.